NEXA
TECHNOLOGY FOR A BRIGHTER TOMORROW

USER GUIDE
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NexaStation AI Workstation
User Guide

High-Performance Desktop for AI, ML and Professional Workloads

- Built for AI and ML
- Workstation-Class Performance
- Expandable Architecture
- Enterprise-Grade Reliability



Designed for Today's AI Innovators

NexaStation AI Workstation delivers the power, expandability and reliability required for local AI model development, machine learning, data science, 3D rendering and professional workloads.

Intel® Core™ i9 14th Gen	NVIDIA® RTX™ 5090	64 GB DDR5	2 TB NVMe SSD
(24 Cores, 32 Threads)	32 GB GDDR7 (AI & Creator Ready)	(up to 256 GB)	(up to 8 TB)

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1. Product Overview

NexaStation AI Workstation is a tower-class professional system designed for local AI development, machine learning, data science, GPU-accelerated engineering and creator workloads. It emphasizes expandability, sustained performance, large memory capacity and multi-drive storage.

Base workstation configuration

Processor	Intel Core i9-class desktop processor, 24 cores / 32 threads
Graphics	NVIDIA RTX 5090-class GPU, 32 GB GDDR7
Memory	64 GB DDR5 ECC-capable platform; expandable to 256 GB
Storage	2 TB NVMe system drive; multiple NVMe/SATA expansion bays; up to 8 TB validated configuration
Networking	2.5 GbE, Wi-Fi 7, Bluetooth 5.4
Expansion	PCIe expansion for additional qualified accelerators / high-speed I/O
Power	High-efficiency 1200 W PSU
Cooling	High-airflow multi-zone cooling
OS	Windows 11 Pro; Linux compatibility depends on component/driver support

2. Intended AI Workloads

Workload	Assessment
Local LLM inference	Excellent; large VRAM supports more demanding local experime
RAG / embeddings	Excellent
Computer vision	Excellent
Model fine-tuning experiments	Strong development platform; production training scale may requ
Data science	Excellent
3D / rendering / creator	Excellent
Multi-container AI services	Excellent

3. Expansion Architecture

The workstation is designed to evolve. Memory, NVMe storage and qualified PCIe devices can be expanded without replacing the entire system. Expansion planning must account for power, thermals, slot clearance and supported drivers.

4. Local AI Environment

Layer	Typical role
GPU driver / compute runtime	Hardware acceleration
Local model runtime	Inference and model management
Python / .NET SDKs	Application and AI engineering
Docker	Containerized AI services
Kubernetes tooling	Local orchestration experiments
Vector database	Local RAG development
Observability	Logs, traces and performance monitoring

5. Operational Guidance

AI workloads can produce sustained heat and power draw. Keep front and rear ventilation unobstructed, use a properly rated power source and monitor temperatures during long-running workloads. Back up model assets, source code and datasets before hardware or firmware changes.

6. Choosing Workstation vs Laptop

Requirement	Recommended
Maximum portability	NexaBook Developer 14
Mobile CUDA / creator work	NexaBook AI Studio 16
Maximum local GPU memory and expansion	NexaStation AI Workstation
Large memory / multi-drive projects	NexaStation AI Workstation
Frequent travel	NexaBook family

Educational product notice

NexaStation AI Workstation is a fictional product created for the NexaStore educational case study. Specifications are intentionally controlled for RAG and agent demonstrations.